Stony Brook Institute at Anhui University

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Chapter 6
Parameter estimation

General concepts of point estimation

- A point estimator T(X) is a statistic intended for estimating a parameter.
- An estimator T(X) is a function of the sample, while an estimate is the realized value of an estimator that is obtained when a sample is actually taken.
- For estimating a parameter θ, usually we denote the estimator as $\hat{\theta}$.
- A point estimate T(x) is an observed value of the estimator.

General concepts of point estimation

- The two most commonly used are: method of Moments and method of Maximum Likelihood Estimation (MLE).

Method of moments estimation

Idea : **Sample moment** $\xrightarrow{\text{Estimate}}$ **Population moment**

- sample moment: $A_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k$
- population moment: $\alpha_k = E(X^k)$

➢ **method:** let $\hat{\alpha}_k = A_k$, which means

$$\hat{\mu} = \widehat{EX} = \bar{X},$$

$$\frac{1}{n}\sum_{i=1}^{n} X_i^2 = \widehat{EX^2} = \widehat{Var(X)} + \widehat{(EX)}^2$$

Method of moments estimation

Specific steps of moments estimation method

(1) Find the first k moments of population X

If the population x is a discrete random variable, its distribution law is:

$$P(x;\theta_1,\theta_2,\cdots,\theta_k)$$

$$E(X^i)=\sum_{l=1}^{n} x_l^{\,i}\cdot P(\theta_1,\theta_2,\cdots,\theta_k),\qquad i=1,2,\cdots,k$$

Method of moments estimation

Specific steps of moments estimation method

(1) Find the first k moments of population X

If population x is a continuous random variable, its density function is:

$$f(x;\theta_1,\theta_2,\cdots,\theta_k)$$

$$E(X^i)=\int_{-\infty}^{+\infty} x^i \cdot f(x;\theta_1,\theta_2,\cdots,\theta_k)\,dx$$

$$i=1,2,\cdots,k$$

Method of moments estimation

In short, $\mu_i = E(X^i)$ is a function of parameters $\theta_1, \theta_2, \cdots, \theta_k$

denote as:

$$\begin{cases} \mu_1 = \mu_1(\theta_1, \theta_2, \cdots, \theta_k) \\ \mu_2 = \mu_2(\theta_1, \theta_2, \cdots, \theta_k) \\ \cdots\cdots\cdots\cdots\cdots\cdots \\ \mu_k = \mu_k(\theta_1, \theta_2, \cdots, \theta_k) \end{cases} \qquad (*)$$

Method of moments estimation

(2) From （*）, the value of $\theta_1, \theta_2, \cdots, \theta_k$ is:

$$\begin{cases} \theta_1 = \theta_1(\mu_1, \mu_2, \cdots, \mu_k) \\ \theta_2 = \theta_2(\mu_1, \mu_2, \cdots, \mu_k) \\ \cdots\cdots\cdots\cdots\cdots\cdots \\ \theta_k = \theta_k(\mu_1, \mu_2, \cdots, \mu_k) \end{cases} \qquad (**)$$

(3) Replace μ_i in (* *) with the estimator $M_i = \frac{1}{n}\sum_{l=1}^{n} X_l^{\,i}$,

and the moment estimator of θ_i is $\hat{\theta}_i$:

$$\hat{\theta}_i = \hat{\theta}_i(M_1, M_2, \cdots, M_k), \quad i = 1, 2, \cdots, k$$

Method of moments estimation

Example 1: Suppose $X \sim B(n,p)$,find $\hat{p}$.

Solution :

since $X \sim B(n,p)$, then $E(X) = np$,

so $p = \frac{1}{n} E(X)$

Therefore, $\hat{p} = \frac{1}{n} \bar{X}$.

Method of moments estimation

Example 2:

Suppose Normal distribution population X with unknown mean μ and variance σ^2. Find the moment estimate of μ, σ^2.

Solution:

Since $E(X) = \mu$, $Var(X) = \sigma^2$,

Thus ,

$$\hat{\mu} = \widehat{EX} = \overline{X},$$

$$\widehat{\sigma^2} = \widehat{Var(X)} = \widehat{EX^2} - \widehat{(EX)}^2$$

$$= \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \overline{X}^2$$

$$= \frac{n-1}{n} S^2$$

Method of moments estimation

Example 3:

Suppose Normal distribution population X with unknown mean μ and variance $\sigma^2 > 0$, $X_1, X_2, \cdots X_n$ is a sample of X.

(1) Find the moment estimator of μ, and σ^2.

(2) When the population (bulb life) obeys $X \sim N(\mu, \sigma^2)$ with parameters unknown, now take 4 bulbs and their service life (hours) were measured as follows:

1502, 1453, 1367, 1650

Find the moment estimator of the parameters.

Method of moments estimation

Solution :

(1) $E(X)=\mu \qquad E(X^2)=D(X)+[E(X)]^2=\sigma^2+\mu^2$

Let
$$\begin{cases} E(X)=\dfrac{1}{n}\sum_{i=1}^{n}X_i \\ E(X^2)=\dfrac{1}{n}\sum_{i=1}^{n}X_i^2 \end{cases}$$

Thus
$$\begin{cases} \mu=\dfrac{1}{n}\sum_{i=1}^{n}X_i \\ \sigma^2+\mu^2=\dfrac{1}{n}\sum_{i=1}^{n}X_i^2 \end{cases} \Rightarrow \begin{cases} \hat{\mu}=\dfrac{1}{n}\sum_{i=1}^{n}X_i \\ \hat{\sigma}^2=\dfrac{1}{n}\sum_{i=1}^{n}X_i^2-(\hat{\mu})^2 \end{cases}$$

Method of moments estimation

Thus, the moment estimator of μ, σ^2 is:

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \left(\frac{1}{n}\sum_{i=1}^{n} X_i\right)^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

Note :

No matter what distribution the population obeys, the expression of the moment estimator of the population mean and variance is the same.

Method of moments estimation

(2). $\because \; X \sim N(\mu, \sigma^2)$

$$\overline{X} = \frac{1}{4}(1502 + 1453 + 1367 + 1650) = 1493$$

$$\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2 = \frac{1}{4}[(1502-1493)^2 + (1453-1493)^2 + (1367-1493)^2 + (1650-1493)^2]$$

$$= 10551$$

So ,the distribution of moment estimates of the mean and variance of a certain bulb life is:

$$\hat{\mu} = 1493, \quad \hat{\sigma}^2 = 10551$$

Method of moments estimation

Example 4. Suppose $X_1, X_2, \ldots X_n$ is a sample from population X, and its pdf is as follows. where θ, μ are unknown parameter, $\theta > 0$.

$$f(x) = \begin{cases} \dfrac{1}{\theta} e^{-(x-\mu)/\theta} & x \geq \mu \\ 0 & \text{others} \end{cases}$$

Find the moment estimator of θ, μ .

Solution :

$$E(X-\mu) = \theta, \quad D(X-\mu) = \theta^2$$

Method of moments estimation

$$E(X)=\mu+\theta, \quad D(X)=\theta^2$$

Let
$$\begin{cases} \mu+\theta=\bar{X} \\ \theta^2=\dfrac{1}{n}\sum_{i=1}^{n}(X_i-\bar{X})^2 \end{cases}$$

$$\Rightarrow \begin{cases} \hat{\mu}=\bar{X}-\sqrt{\dfrac{1}{n}\sum_{i=1}^{n}(X_i-\bar{X})^2} \\ \hat{\theta}=\sqrt{\dfrac{1}{n}\sum_{i=1}^{n}(X_i-\bar{X})^2} \end{cases}$$

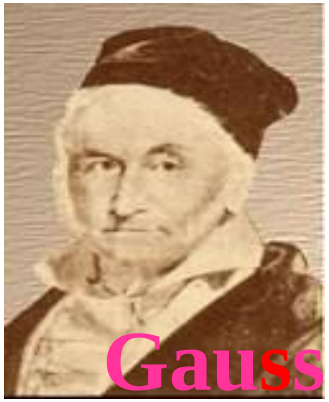
Method of moments estimation

Advantages and Disadvantages of Moments estimation Method

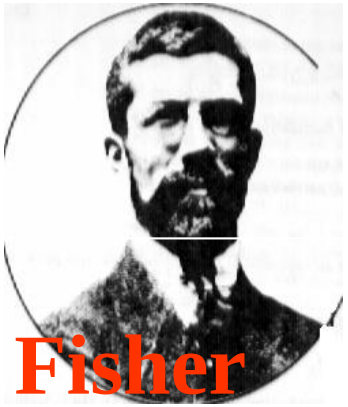
- Advantages: simple
- Disadvantages:
  - For the same parameter to be estimated, different moment equations may give different estimation results;
  - If the total moment does not exist, this method cannot be used

6.2 Maximum Likelihood Estimation -MLE

- **The maximum likelihood method is a kind of parameter estimation method when the population type is known**
- **It was first proposed by Gauss in 1821.**



- **However, this method is often attributed to the British statistician Fischer, mainly because he re proposed and proved some properties of this method in 1912.**



Rationale:

Events occurring in one test should have a high probability of occurrence among all possible results.

6.2 Maximum Likelihood Estimation -MLE

Consider a sample $X_1, X_2, \ldots, X_n$ from a population with probability function $f(x;\theta)=f(x;\theta_1,\ldots,\theta_k)$. Observed values of this sample is denoted by $x_1, x_2, \ldots, x_n$. Then the joint distribution of $X_1, X_2, \ldots, X_n$ has joint probability function

$$f(x_1,\theta)f(x_2,\theta)\cdots f(x_n,\theta)= L(x_1, x_2, \ldots, x_n;\theta)$$

1. fix θ, L is a function sample $x_1, x_2, \ldots, x_n$

2. fix sample $x_1, x_2, \ldots, x_n$, L is a function of paramter θ which is call the likelihood function of the observed values $x_1, x_2, \ldots, x_n$.

6.2 Solution steps of MLE

1. Get likelihood function L(θ)
2. If L is differentiable on θ , get lnL(θ) and the likelihood equations for i=1,2,...,m as follows:

$$\left.\frac{\partial \ln L(\theta)}{\partial \theta_i}\right|_{\theta=\hat{\theta}} = 0$$

3.Solute the likelihood function and get the MLE

$$\hat{\theta}_i = \hat{\theta}_i\left(X_1, X_2, \ldots, X_n\right)$$

4. If L is not differentiable on θ , the MLE must get from the defintion.

Example of MLE

Example 1: If $X_1, X_2, \ldots, X_n$ is a random sample from population B(1,p), find the mle of parameter p.

Solution: Suppose $x_1, x_2, \ldots, x_n$ is the vaule of sample of $X_1, X_2, \ldots, X_n$.

1. the pf of X is

$$P\{X = x\} = p^x (1-p)^{1-x}, \; x = 0,1$$

2. the likelihood function is

$$\boldsymbol{L(p) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i} = p^{\sum_{i=1}^{n} x_i}(1-p)^{n-\sum_{i=1}^{n} x_i},}$$

L(p) is differentiable on p.

Example of MLE

$$\ln L(p) = \left(\sum_{i=1}^{n} x_i\right)\ln p + \left(n - \sum_{i=1}^{n} x_i\right)\ln(1-p),$$

let

$$\frac{\mathrm{d}}{\mathrm{d}p}\ln L(p) = \frac{\sum_{i=1}^{n} x_i}{p} - \frac{n - \sum_{i=1}^{n} x_i}{1-p} = 0,$$

we get the mle of p is

$$\hat{p} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}.$$

It is the same as moment estimation of p.

Example of MLE

Example 2: If $x_1, x_2, \ldots, x_n$ is a random sample value from a normal population $N(\mu, \sigma^2)$, find the mle of μ and σ^2.

Solution: 1. The pdf of sample X_i is

$$p(x_i;\, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}},$$

2. Likelihood function is

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}},$$

which is differiable on μ and σ^2.

3. $$\ln L(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2,$$

Example of MLE

$$\ln L(\mu,\sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2 ,$$

$$let \quad \begin{cases} \dfrac{\partial}{\partial\mu}\ln L(\mu,\sigma^2) = 0 \\ \dfrac{\partial}{\partial\sigma^2}\ln L(\mu,\sigma^2) = 0 \end{cases} \Rightarrow \begin{cases} \dfrac{1}{\sigma^2}\left[\sum_{i=1}^{n} x_i - n\mu\right] = 0 \\ -\dfrac{n}{2\sigma^2} + \dfrac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(x_i-\mu)^2 = 0 , \end{cases}$$

so we get

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{n-1}{n}S^2.$$

Example of MLE

Suppose $X_1, X_2 \cdots X_n$ is a random sample from a normal population with unknown parameters.

Find the MLE of $P(\bar{X} < t)$.

Solution

$$\because \quad X_i \sim N(\mu, \sigma^2), \quad \mu,\ \sigma^2 \text{ unknown.}$$

$$\therefore \quad \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \ \sim \ N(\mu, \frac{\sigma^2}{n})$$

$$P(\bar{X} < t) = P(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{t - \mu}{\sigma/\sqrt{n}}) = \Phi(\frac{t - \mu}{\sigma/\sqrt{n}})$$

Example of MLE

It can be seen from the previous example that:

the MLE of μ is $\bar{X}$,

and the MLE of σ^2 is $\frac{1}{n}\sum_{i=1}^{n}(X_i-\bar{X})^2$.

So, the MLE of $P(\bar{X}<t)$ is $\Phi\left(\frac{t-\bar{X}}{\hat{\sigma}}\sqrt{n}\right)$

where $\hat{\sigma}=\sqrt{\frac{1}{n}\sum_{i=1}^{n}(X_i-\bar{X})^2}$

Example of MLE

Example. Suppose $X_1, X_2, \ldots X_n$ is a sample from population X, and its pdf is as follows, and $\theta > 0$.

$$f(x) = \begin{cases} \theta x^{\theta-1}, & 0 < x < 1 \\ 0, & \text{others} \end{cases}$$

Find the MLE of θ.

Solution :

$$L(\theta) = \prod_{i=1}^{n} \theta x_i^{\theta-1} = \theta^n \left(\prod_{i=1}^{n} x_i\right)^{\theta-1}$$

$(0 < x_i < 1)$

$1 \le i \le n$

Example of MLE

$$L(\theta)=\prod_{i=1}^{n}\theta\, x_i^{\theta-1}=\theta^n\left(\prod_{i=1}^{n}x_i\right)^{\theta-1}$$

$(0<x_i<1)$

$1\le i\le n$

$$\ln L(\theta)=n\ln\theta+(\theta-1)\sum_{i=1}^{n}\ln x_i$$

$$\frac{d\ln L(\theta)}{d\theta}=\frac{n}{\theta}+\sum_{i=1}^{n}\ln x_i=0\,,$$

$$\hat{\theta}=-n\Big/\sum_{i=1}^{n}\ln x_i$$

Example of MLE

Example 3: If $x_1, x_2, \ldots, x_n$ is a random sample value from a uniform population $U(0,\theta)$, find the mle of θ.

Solution: 1. The pdf of Sample X_i is

$$f(x_i \mid \theta) = \begin{cases} \dfrac{1}{\theta}, & 0 \le x_i \le \theta \\ 0, & else \end{cases}$$

2. Likelihood function is

$$L(\theta) = \prod_{i=1}^{n} f(x_i) = \theta^{-n} \qquad 0 < x_i < \theta,$$

which is undifferiable on θ. So we can only find the mle of θ from the defintion of MLE.

Mestimation

3. From the pdf of X, we know when $\theta > x_i$, the $L(\theta) \neq 0$, so from the definition of MLE, the MLE of θ is

$$\hat{\theta}_{MLE} = X_{(n)} = \max(X_1, X_2, \cdots, X_n),$$

and the $L(\theta)=\theta^{-n} \neq 0$.